

William Curran

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Portfolio: <https://williamcurran.github.io/>

Education

University of Massachusetts Lowell - Lowell, MA

Bachelor of Science in Mechanical Engineering, Expected May 2028

GPA: 3.5

Relevant Coursework: Computer-Aided Design, Manufacturing Lab, Materials Science for Engineers, Intro to Mechanical Engineering, Thermodynamics, Statics, Calculus I, II, & III, Physics I & III.

Projects

Vision-Guided Robotic Manipulator (4-DOF Hand Tracking Arm)

- Designed and built a 4-DOF robotic arm controlled through real-time computer vision hand tracking
- Implemented MediaPipe-based hand landmark detection in Python to generate positional control data
- Developed a Python → Arduino control pipeline to translate vision input into MG996R servo motion across multiple joints
- Designed and fabricated custom structural components using SolidWorks and 3D printing for full arm assembly
- Iteratively redesigned arm geometry and link lengths to reduce joint wobble and improve mechanical stability under load

Motion-Detecting Turret (Hand-Tracking System) | Winter 2026

- Designed and built a motion-tracking turret using an Arduino, dual servo motors, and a camera-based vision system
- Implemented real-time hand-tracking in Python using computer vision libraries to generate X- and Y-axis positional data
- Mapped camera coordinates to servo angles (0–180°) for smooth, responsive pan-tilt motion
- Integrated mechanical mounts and custom 3D-printed components to create a fully functional electromechanical system

CNC End-Effector | Fall 2024

- Designed and modeled an end-effector for a CNC robotic arm in SolidWorks to automate object manipulation.
- Programmed machine movement in MATLAB, integrating motion control for precision placement.
- Applied mechanical design principles and manufacturing constraints in project execution.

Technical Skills

CAD & Design: SolidWorks (advanced part and assembly modeling), GD&T

Analysis & Simulation: ANSYS (structural FEA), MATLAB (engineering analysis)

Manufacturing & Prototyping: CNC & Lathe machining, FDM 3D printing, design for assembly

Work Experience

Swim Instructor | JCC Camp Grossman | Summer 2023 - 2025

- Taught students ages 4-13 in structured lessons, adapting instruction to various skill levels.
- Maintained safety and discipline in dynamic environments, demonstrating attentiveness and accountability.
- Built communication and leadership skills through one-on-one and group instruction.

Lifeguard & Receptionist | Dedham Pool | Fall 2022 - Spring 2023

- Monitored pool activity and enforced safety standards, ensuring a secure environment.
- Handled customer service inquiries and scheduling, strengthening multitasking and organizational skills.